

ATIVIDADE AULA_004

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RA: 24190450-2

Exercício-1

```
static final String ENTRADA = "resultado = somaAnterior - valor / 450;"; 2
```

h Main x

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```
"C:\Program Files\Java\jdk-11\bin\java.exe" "-javaagent:C:\Program Files\JetBrai
ALUNO: Natália Possani de Lima | RA: 241904502
ENTRADA: resultado = somaAnterior - valor / 450;
```

LEXEMA	TOKEN

resultado	IDENTIFICADOR
=	ATRIBUICAO
somaAnterior	IDENTIFICADOR
-	SUBTRACAO
valor	IDENTIFICADOR
/	DIVISAO
450	INTEIRO
;	PONTO_E_VIRGULA

Process finished with exit code 0

Exercício-2

```
=== TESTE C2 - identificador personalizado ===
```

```
ENTRADA: int nota_np = 10;
```

LEXEMA	CLASSE	TOKEN
int	LETRA	PALAVRA_RESERVADA
nota_np	LETRA	IDENTIFICADOR
=	SIMBOLO	ATRIBUICAO
10	DIGITO	INTEIRO
;	SIMBOLO	PONTO_E_VIRGULA


```
=== TESTE D - caractere inválido ===
```

```
ENTRADA: int valor# = 1;
```

LEXEMA	CLASSE	TOKEN
int	LETRA	PALAVRA_RESERVADA
valor	LETRA	IDENTIFICADOR
#	DESCONHECIDO	ERRO_LEXICO
=	SIMBOLO	ATRIBUICAO
1	DIGITO	INTEIRO
;	SIMBOLO	PONTO_E_VIRGULA

Exercício-3

The screenshot shows an IDE with a Java file named `Main.java`. The code contains several errors highlighted by the IDE:

- Line 8: `int total = 10 + 20;` - Error: `java: ';' expected` (indicated by a red squiggly line under the semicolon).
- Line 10: `if (total > 20 {` - Error: `java: ')' expected` (indicated by a red squiggly line under the opening curly brace).
- Line 11: `System.out.println("Programa corrigido!);` - Error: `java: ')' expected` (indicated by a red squiggly line under the closing parenthesis).
- Line 12: `System.out.println("Resultado: " + total);` - Error: `java: unclosed string literal` (indicated by a red squiggly line under the opening double quote).

The Build Output window at the bottom shows the following message:

```
03-diagnostics: build failed At 01/09/2026 21:42 sec, 359 ms
Main.java: 3 errors
- ';' expected :8
- ')' expected :10
- unclosed string literal :11
```

The screenshot shows the IntelliJ IDEA interface with a Java file named `Main.java`. The code defines a `Main` class with static variables `ALUNO` and `RA`, and a `main` method. There are three syntax errors highlighted: a missing closing parenthesis on line 10, a missing closing parenthesis on line 11, and a missing closing parenthesis on line 12. The Run window shows the output of the program, which prints the student's name, RA, and the result of the calculation `10 + 20`.

```
1 public class Main {
2     static final String ALUNO = "Natália Possani de Lima";
3     static final String RA = "24190450-2";
4
5     public static void main(String[] args) {
6         System.out.println("ALUNO: " + ALUNO + " | RA: " + RA);
7
8         int total = 10 + 20;
9
10        if (total > 20) {
11            System.out.println("Programa corrigido!");
12            System.out.println("Resultado: " + total);
13        }
14    }
15 }
```

Run: Main

```
"C:\Program Files\Java\jdk-11\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA Community Edition 2025.2.6.1\lib\idea_rt.jar=S5532" -Dfile.encoding=UTF-8
ALUNO: Natália Possani de Lima | RA: 24190450-2
Programa corrigido!
Resultado: 30

Process finished with exit code 0
```

Exercício-4

The screenshot shows the IntelliJ IDEA interface with a Java file named `Main.java`. The code simulates a parser's lookahead process for the expression `5 + 3 * 4`. It prints the current state of the parser, including the current token, the lookahead token, and the result of the calculation. The output shows the parser correctly identifying the operators and the operands, and calculating the result `5 + 3 * 4 = 17`.

```
ALUNO: Natália Possani de Lima | RA: 241904502
ENTRADA: 5 + 3 * 4

ENTRANDO em <expr> | lookahead = 5
ENTRANDO em <term> | lookahead = 5
ENTRANDO em <factor> | lookahead = 5
Numero reconhecido: 5
Novo lookahead: +
SAINDO de <factor> | lookahead = +
SAINDO de <term> | lookahead = +
Encontrado operador +
Novo lookahead: 3
ENTRANDO em <term> | lookahead = 3
ENTRANDO em <factor> | lookahead = 3
Numero reconhecido: 3
Novo lookahead: *
SAINDO de <factor> | lookahead = *
Encontrado operador *
Novo lookahead: 4
ENTRANDO em <factor> | lookahead = 4
Numero reconhecido: 4
Novo lookahead: $FIM$
SAINDO de <factor> | lookahead = $FIM$
SAINDO de <term> | lookahead = $FIM$
SAINDO de <expr> | lookahead = $FIM$
```

Exercício-5

ALUNO: Natália Possani de Lima | RA: 241904502

ENTRADA: id+id*id

PILHA	ENTRADA	ACAO

\$ 0	id+id*id\$	SHIFT id -> estado 5
\$ 0 id 5	+id*id\$	REDUCE F -> id
\$ 0 F 3	+id*id\$	REDUCE T -> F
\$ 0 T 2	+id*id\$	REDUCE E -> T
\$ 0 E 1	+id*id\$	SHIFT + -> estado 6
\$ 0 E 1 + 6	id*id\$	SHIFT id -> estado 5
\$ 0 E 1 + 6 id 5	*id\$	REDUCE F -> id
\$ 0 E 1 + 6 F 3	*id\$	REDUCE T -> F
\$ 0 E 1 + 6 T 9	*id\$	SHIFT * -> estado 7
\$ 0 E 1 + 6 T 9 * 7	id\$	SHIFT id -> estado 5
\$ 0 E 1 + 6 T 9 * 7 id 5	\$	REDUCE F -> id
\$ 0 E 1 + 6 T 9 * 7 F 10	\$	REDUCE T -> T * F
\$ 0 E 1 + 6 T 9	\$	REDUCE E -> E + T
\$ 0 E 1	\$	ACCEPT